1. **The function** 1 point

$$\beta(\mathbf{x},\mathbf{y}) = \mathbf{x}^T \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \mathbf{y}$$

is

- [x] bilinear
- [ ] not bilinear
- [ ] not positive definite
- [ ] not symmetric
- [x] positive definite
- [x] an inner product
- [ ] not an inner product
- [x] symmetric

2. **The function** 1 point

$$\beta(\mathbf{x},\mathbf{y}) = \mathbf{x}^T \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \mathbf{y}$$

is

- [ ] an inner product
- [ ] positive definite
- [x] symmetric
- [ ] not symmetric
- [x] not positive definite
- [ ] not bilinear
- [x] not an inner product
- [x] bilinear

3. **The function** 1 point

$$\beta(\mathbf{x},\mathbf{y}) = \mathbf{x}^T \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \mathbf{y}$$

is

- [ ] symmetric
- [x] not symmetric
- [x] bilinear
- [ ] not bilinear
- [ ] an inner product
- [x] not an inner product

4. **The function** 1 point

$$\beta(\mathbf{x},\mathbf{y}) = \mathbf{x}^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{y}$$

is

- [ ] not symmetric
- [ ] not bilinear
- [x] bilinear
- [x] positive definite
- [ ] not positive definite
- [x] symmetric
- [ ] not an inner product
- [x] an inner product

5. **For any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ write a short piece of code that defines a valid inner product.** 1 point

```python
import numpy as np

```

```python
def dot(a, b):
  """Compute dot product between a and b.
  Args:
    a, b: (2,) ndarray as R^2 vectors

  Returns:
    a number which is the dot product between a, b
  """

  dot_product = a[0]*b[0] + a[1]*b[1]

  return dot_product

# Test your code before you submit.
a = np.array([1,0])
b = np.array([0,1])
print(dot(a,b))
```